

Image Segmentation with K-Means

Objective

The main objective of the project is to segment the given image (which you can find [here](#)) into multiple regions based on the similarity of pixel values in the image. This process is called image segmentation and is an essential step in computer vision and image processing.

The K-means algorithm will be used to divide the image into k clusters, where k is a user-defined parameter. Each cluster will represent a region in the image. The student will then colour each pixel with the colour of the centroid of the cluster they belong to, thus producing a segmented image.

Pre-processing

The image needs to be pre-processed in order to convert it into a suitable format for the K-Means algorithm. This can be achieved using the Pillow Python package, which provides functions for reading, modifying and saving images. You need to convert the image into a 2-dimensional numpy array, where each row represents a pixel and each column represents a colour value (such as red, green and blue).

Deliverables

The following deliverables are expected from the project:

- A Python code that implements the K-Means algorithm for image segmentation, including the pre-processing steps using the Pillow package.
- A report explaining the implementation of the K-Means algorithm for image segmentation, including the pre-processing steps and the results obtained. The report should detail exactly what the student did and how they did it.
- An image with the segmentation applied as described above. The original image and the segmented image should be clearly labeled.
- A discussion on the limitations and potential improvements of the implemented K-Means algorithm for image segmentation.